

Cogging Torque and Acoustic Noise Reduction in Permanent Magnet Motors by Teeth Pairing

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Abstract—This paper investigates reduction of cogging torque and acoustic noise in permanent magnet motors with larger stator slot opening. Using energy method with Fourier series expansion, cogging torque is analytically determined with airgap MMF function and airgap permeance function. It shows that there exist several tooth widths which minimize the cogging torque. It also proposes a teeth pairing with two different types of tooth width which can effectively reduce the cogging torque and thus the acoustic noise. Experimental results show that the proposed teeth pairing reduces the cogging torque by 85% and the acoustic noise by 3.1 dB.

Index Terms—Acoustic noise, cogging torque, permanent magnet motors, teeth pairing.

I. INTRODUCTION

COGGING torque is produced by the interaction between permanent magnets (PM's) and slotted iron structure, and manifests itself by the tendency of a rotor to align in a number of stable positions even when a motor is not energized. It always produces a pulsating torque ripple resulting in vibration and acoustic noise which is detrimental to the motor performance. It is a vital design consideration in electric power steering, home appliances, robotics, machine spindle, high-precision position control and constant speed control, in which accurate motion control is required.

With the advent of high energy magnetic material, smaller but powerful PM motors are realizable due to the increased residual magnetic flux density. However, torque pulsation by cogging torque which is proportional to the square of the flux density aggravates the overall motor performance. Analytical calculation of cogging torque has been found using airgap MMF function [1] and airgap permeance function [2]. As for the design technique, stator teeth notching is proposed to reduce cogging torque using FEM [3]. This paper shows that stator slot width or tooth width is greatly related to the cogging torque variation and the number of tooth widths with minimum cogging torque can be determined by the number of poles and slots. It also shows that teeth pairings with two different types of tooth width can effectively reduce the cogging torque for a motor with a larger stator slot width. Experimental results are also given which support this method for the reduction of cogging torque and acoustic noise.

II. ANALYSIS OF COGGING TORQUE

Cogging torque is energy variation within a motor as the rotor rotates. Since the energy change in PM's and iron is negligible compared to that of air, the magnetostatic energy is given as

$$W(\alpha) \approx W(\alpha)_{\text{airgap}} = \frac{1}{2\mu_0} \int_V P^2(\theta) \cdot F_m^2(\theta, \alpha) dV \quad (1)$$

where μ_0 , $P(\theta)$, $F_m(\theta, \alpha)$ and α are the permeability of air, airgap permeance function, airgap MMF function and rotational angle of rotor, respectively. For motors with constant airgap, airgap permeance function and airgap MMF function are expressed as

$$P(\theta) = \frac{\mu_0}{g} G(\theta) \quad (2)$$

$$F_m(\theta, \alpha) = \frac{g}{\mu_0} B(\theta, \alpha) \quad (3)$$

where $G(\theta)$, $B(\theta, \alpha)$ and g are relative airgap permeance function, flux density function and airgap length. Substituting (2) and (3) to (1), and integrating along the stack length, energy within the airgap can be expressed as

$$W(\alpha)_{\text{airgap}} = \frac{1}{2\mu_0} \cdot L_S \cdot \frac{1}{2} (R_2^2 - R_1^2) \int_0^{2\pi} G^2(\theta) \cdot B^2(\theta, \alpha) d\theta \quad (4)$$

For the external rotor motor, $R_1 = R_s$ and $R_2 = R_m$, and for the internal rotor motor, $R_1 = R_m$ and $R_2 = R_s$, where L_s , R_m and R_s are stack length, PM radius and stator radius, respectively. $G^2(\theta)$ and $B^2(\theta, \alpha)$ can be generally expanded using Fourier series

$$G^2(\theta) = \sum_{n=0}^{\infty} [G_{a_{nN_S}} \cos nN_S\theta + G_{b_{nN_S}} \sin nN_S\theta] \quad (5)$$

$$B^2(\theta, \alpha) = \sum_{n=0}^{\infty} [B_{a_{nN_P}} \cos nN_P(\theta + \alpha) + B_{b_{nN_P}} \sin nN_P(\theta + \alpha)] \quad (6)$$

where N_P and N_S are the number of PM's and the number of slots, and $G_{a_{nN_S}}$, $G_{b_{nN_S}}$, $B_{a_{nN_P}}$ and $B_{b_{nN_P}}$ are the corresponding Fourier coefficients. Substituting (5) and (6) in (4) and utilizing the orthogonality of trigonometric functions as in (7), the cogging torque can be expressed as in (8).

For $n \neq m$,

$$\begin{aligned} \int_0^{2\pi} \cos n\theta \cos m\theta d\theta &= 0 \\ \int_0^{2\pi} \cos n\theta \sin m\theta d\theta &= 0 \\ \int_0^{2\pi} \sin n\theta \sin m\theta d\theta &= 0 \end{aligned} \quad (7)$$

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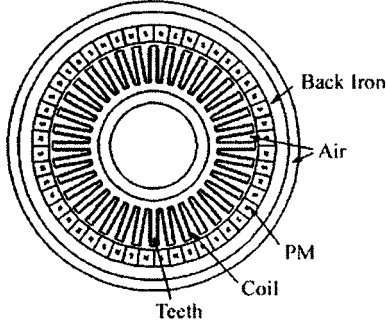


Fig. 1. Schematic of a PM motor to be analyzed.

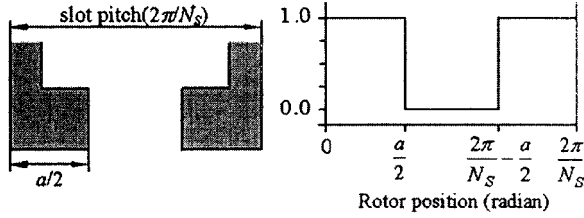


Fig. 2. Simplest airgap permeance function.

$$T(\alpha) = -\frac{\partial W(\alpha)_{airgap}}{\partial \alpha} = \frac{LS\pi}{4\mu_0} (R_2^2 - R_1^2) \cdot \sum_{n=0}^{\infty} nN_L [Ga_{nN_L} Ba_{nN_L} \sin nN_L \alpha - Gb_{nN_L} Ba_{nN_L} \cos nN_L \alpha + Ga_{nN_L} Bb_{nN_L} \cos nN_L \alpha + Gb_{nN_L} Bb_{nN_L} \sin nN_L \alpha] \quad (8)$$

where N_L is the least common multiple of N_P and N_S . For symmetric motors, with all sine terms set to zero in (5) and (6), equation (8) can be simplified as

$$T(\alpha) = \frac{LS\pi}{4\mu_0} (R_2^2 - R_1^2) \cdot \sum_{n=0}^{\infty} nN_L Ga_{nN_L} Ba_{nN_L} \sin nN_L \alpha \quad (9)$$

Equations (8) and (9) reveal that the cogging torque is first governed by N_L , G_{nN_L} and B_{nN_L} with the fundamental period of $2\pi/N_L$. It should be noted that various design techniques can be well explained using the above parameters. Teeth notching, interpole design and teeth pairing are used to reduce G_{N_L} by proper stator design. Magnet shaping and magnet arc design are used to reduce the corresponding B_{nN_L} . For an asymmetric motor such as 8 poles and 9 slots, N_L is increased to reduce the corresponding G_{N_L} and B_{N_L} , thus reducing the cogging torque. Among various design techniques to control N_L , G_{nN_L} and B_{nN_L} , this paper investigates effects of G_{nN_L} by controlling tooth width. Fig. 1 shows the schematic of a 48-pole, 36-slot PM motor to be analyzed. Using the simplest airgap permeance function shown in Fig. 2, one can show that G_{nN_L} can be expressed as

$$G_{nN_L} = \frac{N_S}{\pi} \left[\int_0^{a/2} \cos nN_L \theta d\theta + \int_{2\pi/N_S - a/2}^{2\pi/N_S} \cos nN_L \theta d\theta \right] = \frac{N_S}{\pi} \frac{1}{nN_L} 2(-1)^{n(N_L/N_S)} \sin nN_L \frac{a}{2}. \quad (10)$$

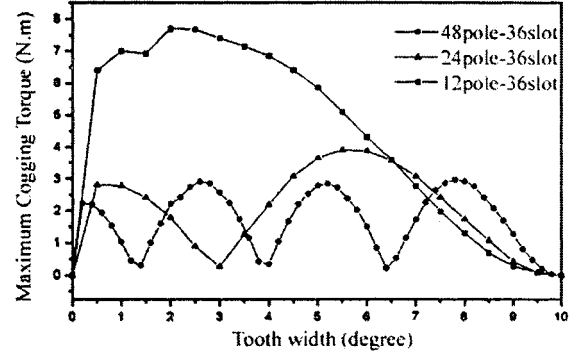


Fig. 3. Effect of tooth width on the maximum cogging torque.

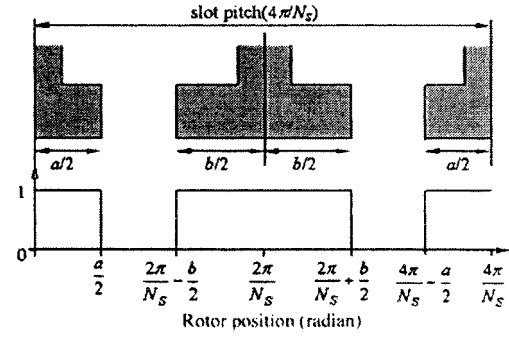


Fig. 4. Simplest airgap permeance function for teeth pairing design.

Noting that the tooth width a can be changed from 0 to $2\pi/N_S$ radians for N_S -slot motors, one can also show that there exist N number of tooth widths that make the cogging torque vanish excluding a slotless stator and N given as

$$N = \frac{N_L}{N_S} - 1. \quad (11)$$

For the prototype motor, the number of tooth widths, N , which makes cogging torque vanish is 3. Fig. 3 shows the effect of tooth width on the maximum cogging torque using FEM analysis and confirms the analysis result in (11) for various types of motors.

III. STATOR TEETH PAIRING DESIGN

Fig. 4 shows a teeth pairing with two different tooth widths a and b , and the corresponding simplest airgap permeance function. Noting that the period of airgap permeance function is doubled, N_L has to be replaced by the least common multiple of $N_S/2$ and N_P . Using the previous analysis, one can also show that the G_{nN_L} with two different tooth widths a and b can be expressed as

$$G_{nN_L} = \frac{N_S}{\pi} \left[\int_0^{a/2} \cos nN_L \theta d\theta + \int_{2\pi/N_S - b/2}^{2\pi/N_S + b/2} \cos nN_L \theta d\theta + \int_{4\pi/N_S - a/2}^{4\pi/N_S} \cos nN_L \theta d\theta \right] = \frac{N_S}{\pi} \frac{1}{nN_L} 2 \left[\sin nN_L \frac{a}{2} + \sin nN_L \frac{b}{2} \right]. \quad (12)$$

From (12), the fundamental component of the cogging torque can be eliminated by appropriate combinations of a and b for

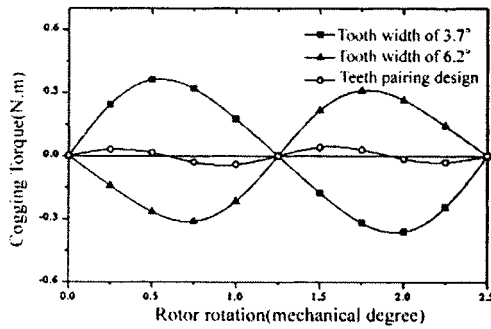


Fig. 5. Cogging torque (FEM).

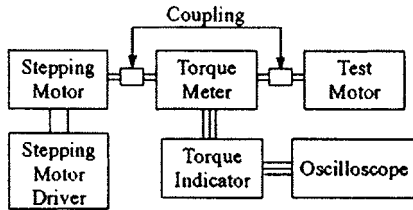


Fig. 6. Experimental setup (cogging torque).

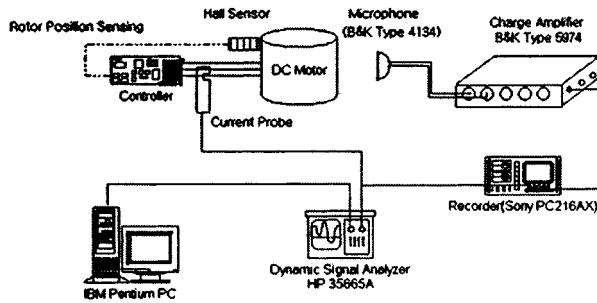


Fig. 7. Experimental setup (acoustic noise).

the case of $n = 1$. It should be noted that those combinations cannot make G_{2N_L} and higher harmonics vanish. Using a teeth pairing design, however, the fundamental component which is dominant in the cogging torque can be effectively eliminated. Fig. 5 shows the FEM result for the teeth pairing design of tooth width of 3.7° and 6.2° . The fundamental frequency of the cogging torque is changed to the double with the remarkable reduction of the magnitude and experimental results in Fig. 8 confirm the analysis. Using a teeth pairing, it can be seen that the maximum cogging torque is reduced by 85% compared to the prototype. Figs. 6 and Fig. 7 show experimental setup to measure cogging torque and acoustic noise. Effect of the cogging torque on acoustic noise is shown in Fig. 9 with the rotational speed of 750 rpm. Noting that the fundamental frequency of the given motor is the 144th harmonic of the rotating frequency, which is the least common multiple of 48 poles and 36 slots, the cogging

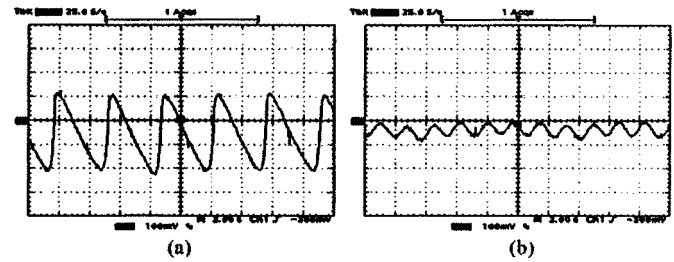


Fig. 8. Cogging torque (experimental) (a) prototype (b) teeth pairing.

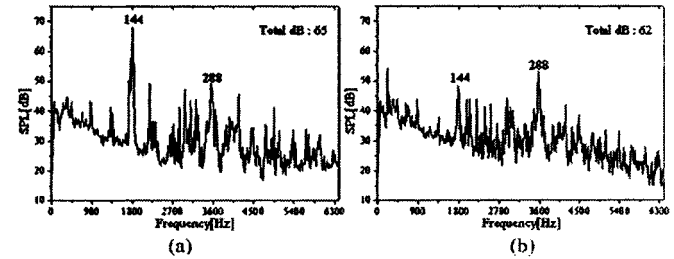


Fig. 9. Acoustic noise (a) prototype (b) teeth pairing.

torque is one of the dominant sources of acoustic noise in PM motors. For the teeth pairing design motor, it can be seen that the fundamental component of the noise spectrum has been substantially reduced due to the reduced cogging torque. Noting that the commutation torque ripple also occurs at the fundamental frequency of 144th harmonic with 3 phase winding, acoustic noise at the fundamental and higher harmonics by torque ripple still remains.

IV. CONCLUSION

This paper analytically calculates cogging torque using the energy method with Fourier series expansion and reveals that N_L , G_{nN_L} and B_{nN_L} are the dominant design parameters to reduce the cogging torque. It also analytically verifies the number of tooth widths that minimizes the cogging torque by controlling G_{nN_L} . As a design technique, teeth pairing is proposed to reduce the cogging torque and the acoustic noise. Experimental results are also followed to confirm the proposed design for both the cogging torque and acoustic noise.

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